CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in LLM-based applications, from RAG and agents to vector search, backend integration and DevOps practices. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning and Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Sole developer owning an LLM-based knowledge-extraction pipeline end-to-end, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized a multi-agent system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents; validated it against an SAP-system-derived ground truth and raised precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated into Claude Code.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts validate and approve knowledge-graph changes.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- **Tech stack: Python, LiteLLM, LangChain, LangGraph, RAG, GraphRAG, Vector Search, MCP, FastAPI, React, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Built and deployed an agentic AI application with LangGraph that routes decision-making across vector database queries, general inference, and summarization, with a React frontend and a Python FastAPI backend whose REST APIs were tested with Postman.
- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated the requirements into a solution approach.
- Containerized the stack with Docker and deployed it to AWS with CDK, with version control and CI/CD in GitLab.
- **Tech stack: Python, LangChain, LangGraph, FastAPI, REST APIs, React, Postman, Docker, GitLab CI/CD, AWS (CDK, EC2, Lambda)**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data, and built retrieval pipelines covering chunking strategies, vector databases, indexing, and fine-tuning models for better retrieval.
- **Tech stack: Python, Hugging Face Transformers, PyTorch, LangChain, RAGAS, PytrecEval, Azure AI Search, AWS CDK**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, with a Streamlit dashboard where chemists simulate parameter changes.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA
- Fine-tuned the open-source Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

Project: Knowledge Graph | NLP (NLTK) | Protégé | SPARQL | RDF/OWL
- Developed a knowledge graph used as a learning resource for AI students with Protégé and rdflib (Python): information extraction with NLTK, RDF transformation, SPARQL querying, and OWL enhancement.

TECHNICAL SKILLSET

- **Large Language Models:** open-source and API-based LLMs (LiteLLM, Hugging Face Transformers), RAG architectures, agents and multi-agent systems, LangChain, LangGraph, prompt engineering, MCP, fine-tuning (QLoRA), RAGAS
- **Vector Databases & Search:** Qdrant, Pinecone, Azure AI Search, similarity search, embedding models
- **Backend & Frontend:** Python, FastAPI, REST APIs, Postman, Python microservices, React, Streamlit
- **Databases:** SQL, Postgres, MongoDB, Neo4j, DynamoDB
- **MLOps & DevOps:** Git, GitLab CI/CD, Docker, Kubernetes, AWS (CDK, Lambda, EC2, SQS, S3), Azure
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, NLTK, time-series analysis
- **Ways of Working:** Agile, Claude Code (custom skills, agentic coding workflows)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)